

VERNAL, UT, USA

WMO: 725705

Lat: 40.443N

Lon: 109.513W

Elev: 1606

StdP: 83.46

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94030

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-17.9	-23.6	0.5	-20.0	-21.1	0.7	-16.3	7.6	-2.1	5.6	-1.9	0.8	280	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	34.1	15.4	32.8	15.1	31.6	14.9	17.8	28.3	16.9	27.7	16.3	27.4	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	12.7	20.2	13.4	11.7	20.4	12.4	10.9	20.5	56.0	27.7	53.5	27.4	51.4	27.2	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.8	6.3	-22.2	36.1	3.6	1.0	-24.8	36.9	-26.9	37.4	-29.0	38.0	-31.6	38.7		
(5)			-22.6	18.8	3.5	0.9	-25.1	19.5	-27.2	20.0	-29.1	20.4	-31.7	21.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-7.3	-2.8	4.3	8.7	13.8	19.6	23.7	21.7	16.7	9.1	1.8	-5.9	(6)	
	DBStd	11.12	5.58	5.21	4.19	3.82	4.14	3.80	2.15	2.21	3.57	4.08	4.21	5.38	(7)	
	HDD10.0	1969	537	359	182	68	14	1	0	0	3	64	248	494	(8)	
	HDD18.3	3898	795	592	436	290	147	31	0	3	69	287	497	752	(9)	
	CDD10.0	1482	0	0	5	28	132	289	424	363	204	36	1	0	(10)	
	CDD18.3	371	0	0	0	0	7	69	166	108	21	0	0	0	(11)	
	CDH23.3	5253	0	0	1	10	219	1156	2127	1331	399	11	0	0	(12)	
	CDH26.7	2251	0	0	0	1	49	491	1054	549	107	1	0	0	(13)	
(14) Wind	WSAvg	2.2	1.2	1.6	2.4	3.1	3.0	2.9	2.4	2.4	2.3	2.1	1.8	1.4	(14)	
(15) Precipitation	PrecAvg	216	12	13	17	20	23	19	14	17	24	30	15	14	(15)	
	PrecMax	307	51	32	56	56	71	58	51	61	73	111	39	49	(16)	
	PrecMin	84	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	56	11	9	14	13	18	17	13	13	18	26	10	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	12.6	21.3	24.6	30.9	35.0	36.3	34.4	31.7	25.0	16.2	8.8	(19)	
		MCWB	2.5	4.6	8.5	10.0	14.2	14.9	16.6	15.5	14.4	11.4	6.9	3.6	(20)	
	2%	DB	4.1	9.6	17.9	22.2	27.8	33.0	34.9	32.8	29.4	22.1	13.7	5.6	(21)	
		MCWB	0.4	3.3	6.9	8.8	12.4	14.1	15.9	15.3	13.7	10.6	6.0	1.3	(22)	
	5%	DB	2.3	7.0	15.7	20.1	26.0	31.6	33.6	31.6	27.6	19.5	11.6	3.1	(23)	
		MCWB	-0.4	2.1	5.9	7.9	11.5	13.8	15.7	15.1	13.4	9.7	5.0	-0.2	(24)	
	10%	DB	0.4	4.4	12.9	17.8	23.6	29.7	32.3	30.0	25.9	17.3	9.0	1.3	(25)	
		MCWB	-1.4	0.7	4.7	7.2	10.7	13.3	15.5	14.9	12.5	8.3	3.8	-1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	5.4	9.0	10.9	15.1	16.4	19.0	18.6	16.9	13.0	8.2	4.0	(27)	
		MCDB	6.0	11.4	19.5	23.1	28.7	30.3	29.7	28.9	26.2	21.9	14.4	8.0	(28)	
	2%	WB	1.3	3.7	7.6	9.6	13.2	15.5	18.0	17.5	15.6	11.6	6.7	1.9	(29)	
		MCDB	3.3	8.2	17.0	19.4	24.8	29.3	28.8	27.3	24.8	18.9	12.0	4.4	(30)	
	5%	WB	0.0	2.5	6.3	8.7	12.3	14.8	17.3	16.8	14.7	10.5	5.5	0.4	(31)	
		MCDB	1.8	6.3	14.3	18.2	23.5	28.3	28.3	26.7	23.5	17.8	10.6	2.8	(32)	
	10%	WB	-1.4	1.1	5.2	7.7	11.3	14.0	16.7	16.2	13.8	9.4	4.2	-0.9	(33)	
		MCDB	0.4	4.1	12.5	16.8	22.1	27.1	28.1	26.1	22.3	16.0	8.6	1.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.9	13.6	14.5	15.4	17.2	16.9	16.5	16.0	14.1	12.0	9.7	(35)	
		MCDBR	11.4	13.2	17.8	18.6	18.9	19.4	18.6	18.8	18.7	16.9	15.1	12.0	(36)	
		MCWBR	8.2	8.2	9.0	8.5	7.6	7.0	6.4	6.7	7.4	8.2	8.7	8.4	(37)	
	5% WB	MCDBR	10.0	11.9	16.5	16.6	17.2	17.6	15.7	15.4	15.2	15.1	13.7	10.7	(38)	
		MCWBR	7.5	7.7	8.6	7.9	7.1	6.8	5.6	5.9	6.4	7.4	8.2	7.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.254	0.265	0.282	0.305	0.317	0.318	0.354	0.357	0.310	0.288	0.262	0.251	(40)	
	taud		2.523	2.530	2.511	2.456	2.477	2.490	2.428	2.417	2.531	2.562	2.573	2.555	(41)	
	Ebn at Noon		933	969	980	970	959	953	915	905	936	927	916	910	(42)	
	Edh at Noon		72	83	95	108	108	107	113	111	92	80	68	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.19	3.12	4.50	5.73	6.70	7.73	7.05	6.29	5.40	3.87	2.62	1.91	(44)	
	RadStd		0.19	0.25	0.31	0.32	0.43	0.49	0.34	0.27	0.34	0.26	0.20	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page